Video-Text Multi-Instance Retrieval in Egocentric Video: From a JPoSE Baseline to AVION ViT-L Fine-Tuned with Symmetric Multi-Similarity Loss on EPIC-KITCHENS-100

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Abstract. Multi-instance video-text retrieval in egocentric video poses challenges that binary contrastive objectives do not capture: a single textual query may align with several video segments at different degrees of relevance, and first-person video introduces strong viewpoint, illumination and motion variability. We address the EPIC-KITCHENS-100 Multi-Instance Retrieval (MIR) challenge through three progressively stronger approaches: (i) a JPoSE-based zero-fine-tuning baseline that exploits part-of-speech embeddings; (ii) a score-level ensemble of three pre-trained retrieval models (JPoSE, MI-MM and MMEN) with optional graph-diffusion re-ranking on visual and textual k-nearest-neighbour graphs; and (iii) end-to-end fine-tuning of an AVION ViT-L/14 dual encoder with the Symmetric Multi-Similarity (SMS) loss, which directly consumes the soft-label relevancy matrices, enhanced with horizontal-flip and longer-clip test-time augmentation. Across six iterations, average nDCG on the official Codabench leaderboard rises from **53.53** (JPoSE alone) to **55.82** (ensemble + re-ranking) and finally to **69.68** (AVION ViT-L + SMS, 16 epochs, with TTA), an absolute gain of **+16.15** nDCG. We analyse where each gain originates, whether encoder capacity, soft-label supervision, test-time augmentation and additional epochs, and outline a model-soup-style ensemble of differently-trained ViT-L+SMS instances as future work, motivated by the compute and offline-evaluation constraints encountered.

Keywords: Egocentric video · Multi-instance retrieval · Vision-language models · Symmetric multi-similarity loss · EPIC-KITCHENS-100.

1 Introduction

The growth of audiovisual content has driven the development of methods capable of analysing, indexing and automatically retrieving visual information. Within this landscape, multimodal video–text retrieval has consolidated as a relevant area at the intersection of computer vision and natural language processing. An especially challenging scenario is that of *egocentric video*, captured from a first-person perspective using head- or body-mounted cameras. Such recordings register natural interactions between hands, objects and the surrounding environment, introducing strong viewpoint, illumination and motion variability. Models must therefore interpret actions and contextual relationships at fine granularity in order to establish precise correspondences between visual content and textual descriptions.

The reference resource for this problem is the EPIC-KITCHENS-100 dataset [2], which contains over 100 h of egocentric kitchen video and approximately 90,000 narrated action segments. The associated Multi-Instance Retrieval (MIR) challenge formalises the task: given a textual query, the system must rank 9,668 trimmed video segments from the test split (and vice versa) by their semantic relevance, where ground-truth relevance is provided as a continuous *soft-label matrix* rather than as one-to-one binary correspondences [2,23]. This formulation acknowledges that several segments may match a query at different degrees of similarity and so, is evaluated with order-sensitive metrics like *normalized discounted cumulative gain* (nDCG) [8] and *mean average precision* (mAP) rather than with the conventional Recall@K used in third-person retrieval. Despite recent progress in multimodal foundation models, semantic alignment between egocentric video and natural language remains difficult due to dynamic perspective changes, occlusions and multi-object interactions hinder the extraction of robust visual representations, and the fact that most existing losses do not exploit the graded relevance signal provided by the challenge.

Contributions. This work systematically explores the EK-100 MIR task through three progressively stronger approaches that share an iterative development methodology grounded on the same evaluation protocol. Concretely, we contribute:

1. A reproduction and analysis of a **JPoSE-based zero-fine-tuning baseline** [24], in which verb, noun and action sub-embeddings of pre-trained Part-of-Speech encoders are concatenated to produce a 9668×3842 similarity matrix. This baseline reaches **53.53 nDCG** on the official Codabench leaderboard and provides the lower bound of our study.
2. A **score-level ensemble** that fuses three pre-trained retrieval models (JPoSE, MI-MM [14] and MMEN [24]) through row-wise min-max normalization followed by equal-weight averaging, optionally followed by a **graph-diffusion re-ranking** step that propagates similarity scores through visual and textual k-nearest-neighbour graphs in the spirit of [7]. Re-ranking provides a further uplift over the ensemble alone, reaching **55.82 nDCG**.
3. A complete **end-to-end fine-tuning pipeline** that trains an AVION ViT-L/14 dual encoder [25] with the Symmetric Multi-Similarity (SMS) loss [19], which consumes the soft-label relevancy matrices directly during training, complemented by a *horizontal-flip + 32-frame* test-time augmentation strategy. The final iteration, trained for sixteen epochs on a single GPU, attains

69.68 nDCG on the official Codabench leaderboard—a +**16.15** absolute gain over the JPoSE baseline.

4. A discussion of where each gain originates, whether encoder capacity, soft-label supervision, test-time augmentation or additional epochs, together with a report of the primary limitations encountered (single-GPU compute budget and unavailability of the test-set relevancy matrix for offline evaluation) and a future-work proposal centered on a *model-soup*-style ensemble [22] of differently-trained ViT-L+SMS instances.

2 Related Work

The line of work that frames our contribution starts from contrastive vision–language pretraining and traces how each successive refinements, such as adaptation to video, specialization to first-person footage, accommodation of graded relevance, and inference-time fusion, reshaped what is achievable on benchmarks like EK-100 MIR. CLIP [16] established the paradigm by showing that contrastive alignment of 400M image–text pairs produces general-purpose visual representations whose zero-shot transfer matches fully supervised baselines, displacing the need to train task-specific encoders from scratch. Bringing the paradigm to video required rethinking how individual frames combine into a clip-level signal: CLIP4Clip [12] compared three temporal aggregation strategies and found that the quality of the pretrained representation outweighs the choice of aggregation, an observation that X-CLIP [13] pushed further with multi-granularity contrastive learning that aligns video–sentence pairs at the global level and frame–word pairs at the local level via a cross-frame communication transformer. ViFi-CLIP [17] closed this trajectory by showing that end-to-end fine-tuning with minimal architectural modifications already achieves competitive zero-shot generalization, evidence that shared embedding spaces transfer well to new domains and motivating our choice of a CLIP-style dual encoder fine-tuned directly with retrieval-aware supervision in Approach 3.

These third-person results, however, do not transfer cleanly to the first-person setting that drives EK-100. Egocentric footage exhibits dynamic camera motion, high intra-class variability and a distribution very different from third-person video. Thus, only at the scale of Ego4D [5], which provides 3,600 h of first-person recordings from 931 participants, did large-scale pretraining for this domain become feasible. EPIC-KITCHENS-100 [2] provided the cooking-domain reference benchmark and, crucially, formalized the multi-instance retrieval task with *soft-label* relevance matrices, evaluated with order-sensitive metrics (nDCG [8], mAP) rather than the binary R@K used in third-person retrieval; the construction of these matrices was later given a principled grounding by Wray et al. [23] via word2vec similarity over verb and noun classes. This data infrastructure enabled a generation of egocentric encoders that progressively closed the gap with third-person performance. First, EgoVLP [10] introduced EgoNCE, an InfoNCE variant that weights additional positives through verb/noun matching that led to improvements over general-domain methods on EK-100 but that still relies on lexical overlap rather than on the official graded matrices. LaViLa [26] introduced a fine-tuned a language model on Ego4D to generate diverse training narrations, becoming the reference video encoder for subsequent work. Subsequently, EgoVLPv2 [15] integrated cross-modal fusion directly into the backbone via interleaved cross-attention. Finally, AVION [25] building on LaViLa's encoder design, trained a ViT-L/14 backbone on a single machine in a single day, won the CVPR 2023 EK-100 challenge and, in combination with FlashAttention [3] for memory-efficient computation, became the de-facto backbone for downstream retrieval work—and, in our pipeline, the visual encoder of Approach 3.

Still, strong encoders alone do not exploit the soft-label structure of the EK-100 MIR task because the loss function decides whether the graded relevance R enters the gradient signal at all. The official challenge baseline reflects this gap, combining JPoSE [24], which learns separate verb, noun and action embedding spaces, with MI-MM, a multi-instance MaxMargin adaptation of MIL-NCE [14]; both operate with binary relevance and ignore the soft-label matrices. JPoSE itself was built on top of the simpler Multi-Modal Embedding Network (MMEN), an MLP-based projection used as a baseline in the same work and one we revisit as the third member of our score-level ensemble. The decisive step beyond binary relevance came with the Symmetric Multi-Similarity (SMS) loss of Wang, Wang and Chau [19]. This loss extends the multi-similarity formulation of [20] so that, instead of binary positive/negative pair weights, the contribution of each pair is modulated by its soft relevance score, applied symmetrically over both modalities. Built on LaViLa as the base encoder, SMS is the most explicit attempt to date to directly optimize the official challenge metric and is the loss we adopt in Approach 3. An adjacent line of research within the broader retrieval community has developed losses that approximate ranking metrics directly. For example, SoDeep [4] learns a sorting surrogate for differentiable mAP and Recall and, Smooth-AP [1] provides a sigmoidal AP approximation. That being said, neither family has been systematically explored in video–text retrieval with graded relevance, leaving a clear gap that our discussion in Section 7.3 returns to. Beyond loss design, scaling has continued to push the limits of multimodal understanding (InternVideo2 [21] reaches state of the art across more than sixty benchmarks via masked video pretraining, contrastive alignment and instruction tuning), and the current EK-100 MIR public leader is CR-CLIP [6], which builds on the AVION dual-encoder architecture and adds a context-refinement module to reach 82.10 nDCG and 66.8 mAP.

Because each of the strands above reshapes only a single component of the retrieval pipeline (the encoder, the loss or the data) there remains a complementary line of work that operates at inference time, and that our Approach 2 explicitly tests. Score-level ensemble fusion combines similarity matrices of complementary models, typically by row-wise normalization followed by weighted averaging, to reduce variance with no additional training cost. Re-ranking refines initial scores by exploiting the local geometry of the retrieval space: Iscen et al. [7] introduced an efficient diffusion procedure on region manifolds in which scores are propagated through a sparse k-nearest-neighbour graph, and k-reciprocal encoding [27] formalizes mutual nearest-neighbor relations as a re-ranking signal; we adapt these ideas to the video–text setting by propagating ensemble scores through both visual and textual k-NN graphs (Eq. (3)). At the parameter level, *model soups* [22] average the weights of several models fine-tuned with different hyperparameters and consistently match or exceed the best individual model with inference cost equal to that of a single one. Wang et al. [19] apply a related strategy in their winning EK-100 MIR submission: a six-model score-level ensemble that mixes Adaptive MI-MM and SMS variants trained at the same length but with different learning-rate schedules and relaxation thresholds. This approach is discussed in Section 7.3 as the natural continuation of our pipeline.

Read together, these four strands expose a structural mismatch with the EK-100 MIR setting that no single one resolves. For one, methods designed for third-person video assume binary relevance and egocentric-specialized encoders such as EgoVLP and LaViLa only partially leverage the soft-label matrices (EgoNCE replaces them with lexical overlap, LaViLa applies standard contrastive learning). Meanwhile, approaches that consume the soft labels directly, such as SMS, do so as a fine-tuning stage on top of binary-pretrained representations and ensemble or re-ranking strategies have rarely been combined explicitly with the soft-label EK-100 setting. Our work bridges these gaps along a single, controlled iteration trajectory: Approach 1 reproduces the binary-relevance baseline, Approach 2 quantifies how far inference-time fusion of frozen soft-label-agnostic models can carry it, and Approach 3 fine-tunes a strong egocentric encoder with relevance-aware supervision on the same evaluation protocol, isolating the contribution of every design choice along the way.

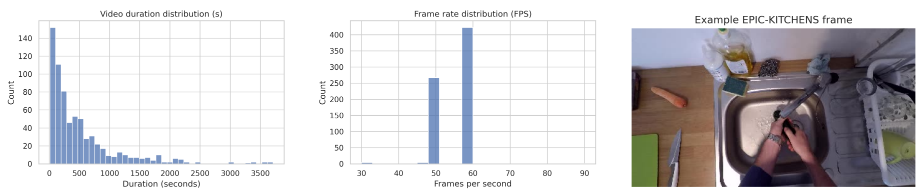
3 Materials

This section describes the EPIC-KITCHENS-100 dataset, the empirical exploration that informed our preprocessing decisions, and the pre-trained checkpoints, hardware and software stack that supported the three approaches.

3.1 Dataset and empirical exploration

We use the EPIC-KITCHENS-100 dataset, focusing on the Multi-Instance Retrieval (MIR) challenge. The corpus comprises trimmed egocentric action segments paired with English narrations, recorded with head-mounted cameras in 45 kitchens and totaling 100 hours of Full-HD video. This encompasses around 20 M frames distributed in 700 variable-length sequences with 97 verb classes and 300 noun classes [2]. The retrieval split releases 67,217 training segments and 9,668 test segments paired with 3,842 unique narrations on the test side. Ground-truth relevance is provided as the soft-label matrix released by Wray et al. [23] alongside the dataset. The *test* relevance matrix is held out by the organizers and available exclusively through the Codabench grader, restricting offline metric computation to the public training and validation splits. Figure 1 illustrates the visual diversity of the corpus.

Fig. 1: Qualitative overview of EPIC-KITCHENS-100 with multiple example frames illustrating different kitchens, viewpoints and activities.



(a) Video duration. (b) Frame rate. (c) Example frame.

Fig. 2: Video-level distributions of duration (s) and frame rate (fps) and a representative training frame illustrating the egocentric viewpoint and typical kitchen clutter.

Video- and image-level statistics. The official `EPIC_100_video_info.csv` reports a mean video duration of ≈ 514 s (std. 599 s) across the 700 recordings, ranging from $\sim$10 s clips to long sessions exceeding one hour, and an average frame rate of 55.9 fps with most recordings at 50 or 59.94 fps. Videos are predominantly Full HD (1920×1080): 694 of 700 recordings use this resolution, yielding an average of $\approx 2.07 \times 10^6$ pixels per RGB frame; appearance statistics computed on a uniform sample of 1,000 training frames yield mean RGB intensities of $\approx (0.44, 0.37, 0.36)$ and per-channel standard deviation ≈ 0.21 on a 0–1 scale, summarizing typical kitchen illumination levels.

Segment-level statistics for MIR. Segment durations are computed from the retrieval annotation files as $\texttt{dur} = (\texttt{stop\_frame} - \texttt{start\_frame} + 1)/\texttt{fps}$. For the 67,217 training segments, the mean is 3.18 s (std. 5.53, median 1.6, IQR $[1.0, 3.18]$ s), confirming that most clips are very short while a long tail of extended actions reaches ~ 297 s. This long tail motivates the choice of clip length used by our AVION-based fine-tuning pipeline (Section 4.4), where short default clips of 8 frames are augmented at test time to 32 frames for richer temporal context.

3.2 Pre-trained checkpoints, hardware and software

Each approach reuses publicly released checkpoints. Approaches 1 and 2 operate on pre-extracted features and were executed on a single NVIDIA T4 (16 GB) provided by Google Colab, using the JPoSE and MMEN encoders distributed by Wray et al. [24] together with the MI-MM checkpoint and S3D-HowTo100M features released alongside MIL-NCE [14]. Approach 3 fine-tunes a ViT-L/14 backbone from the AVION pre-train checkpoint of Zhao and Krähenbühl [25]. It was deployed on a single NVIDIA A100 (40 GB), with gradient checkpointing, the `use-fast-conv1` optimization of AVION and FlashAttention [3] when available, halving attention memory at no accuracy cost, AdamW optimization [11], a base learning rate of 2×10^{-5} and batch size 48. Approach 3 also requires staging $\approx$ 720 GB of EK-100 video chunks (320p, 15 s, 30 fps, libx264) on persistent storage, together with the EK-100 retrieval annotations and the public training relevancy matrix [23]. As mentioned, the test relevancy matrix is held out by the organizers and not used at any stage. The implementation is built on PyTorch 2.4.1 (CUDA 12.1) with the standard vision–language components (`open_clip_torch`, `openai/CLIP`, `transformers`, `timm`), `decord` for video I/O, and `numpy`/`scipy.sparse` for the ensemble and re-ranking of Approach 2. Submissions are packaged as pickle protocol 2 with a numpy namespace compatibility shim required by the Codabench grader. The SMS-Loss code used in Approach 3 is a patched fork of the public release [18,19] that fixes a small number of bugs preventing stable end-to-end runs on EK-100 (tuple unpacking in the gradient-accumulation branch, an undefined `accum_freq` reference replaced by `update_freq`, dataloader retry logic for missing or corrupt video chunks, and `module.`-prefix stripping when loading `DataParallel`-wrapped checkpoints).

4 Methods

We approached EK-100 MIR as an iterative search through the design space, starting from a feature-based zero-fine-tuning baseline and ending with end-to-end fine-tuning of a strong egocentric backbone with soft-label supervision. The three approaches share a common formalization, stated once before each iteration is described.

4.1 Task formalization and common notation

Let $\mathcal{V} = \{v_i\}_{i=1}^{N_v}$ denote the $N_v = 9{,}668$ test video segments and $\mathcal{T} = \{t_j\}_{j=1}^{N_t}$ the $N_t = 3{,}842$ unique text queries on the test split. A retrieval system produces a similarity matrix $S \in \mathbb{R}^{N_v \times N_t}$ such that S_{ij} is large when v_i is semantically relevant to t_j. Ground-truth relevance is encoded in the soft-label matrix $R \in [0,1]^{N_v \times N_t}$ [23] and the official challenge metric is the average normalized discounted cumulative gain

$$\mathrm{nDCG}_{\mathrm{AVG}} = \tfrac{1}{2}\big(\mathrm{nDCG}_{\mathrm{V}\to\mathrm{T}}(S,R) + \mathrm{nDCG}_{\mathrm{T}\to\mathrm{V}}(S^\top, R^\top)\big), \tag{1}$$

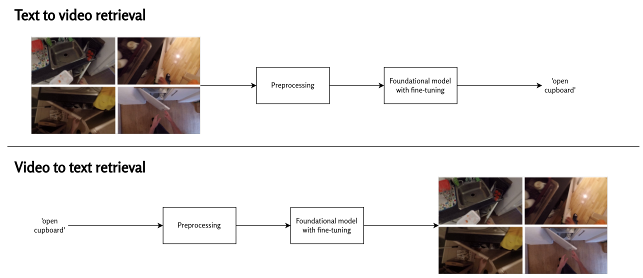


Fig. 3: Overview of the multi-instance retrieval pipeline shared by the three approaches: visual and textual encoders project segments and queries into a shared space, and the resulting similarity matrix S is ranked and evaluated against the soft-label matrix R. Image components adapted from [2].

which captures both retrieval directions on equal footing. The three approaches differ only in how S is constructed while the evaluation protocol is identical across them. Figure 3 summarizes the resulting pipeline.

4.2 Approach 1: JPoSE Part-of-Speech baseline

Our entry point is a feature-based, zero-fine-tuning baseline using JPoSE [24]. It projects segments and queries into verb, noun and full-action sub-spaces, learned jointly with triplet supervision. Let $\phi^{\mathrm{v}}_{\mathrm{V}}, \phi^{\mathrm{v}}_{\mathrm{N}}, \phi^{\mathrm{v}}_{\mathrm{A}}$ and $\phi^{\mathrm{t}}_{\mathrm{V}}, \phi^{\mathrm{t}}_{\mathrm{N}}, \phi^{\mathrm{t}}_{\mathrm{A}}$ denote the visual and textual sub-encoders. We compute joint embeddings as the concatenation $\mathbf{f}_{\mathrm{v}}(v_i) = [\phi^{\mathrm{v}}_{\mathrm{V}}(v_i)\,;\, \phi^{\mathrm{v}}_{\mathrm{N}}(v_i)\,;\, \phi^{\mathrm{v}}_{\mathrm{A}}(v_i)]$ and analogously $\mathbf{f}_{\mathrm{t}}(t_j)$, and obtain $S^{\mathrm{JPoSE}}_{ij} = \cos(\mathbf{f}_{\mathrm{v}}(v_i), \mathbf{f}_{\mathrm{t}}(t_j))$. We use the released `JPoSE_BEST` checkpoint without modification on pre-extracted features. By design this approach cannot exploit R and inherits the limitations of triplet supervision with binary positives and negatives, but it provides a reproducible reference point against which all subsequent iterations are compared.

4.3 Approach 2: score-level ensemble with graph-diffusion re-ranking

The second approach asks how far a purely inference-time strategy can carry the JPoSE baseline by combining several pre-trained retrieval models and refining the result with neighborhood-aware re-ranking. Three complementary models are queried independently: S^{JPoSE} as in Section 4.2; $S^{\mathrm{MI\text{-}MM}}$, the multi-instance MaxMargin model of Miech et al. [14] with S3D-HowTo100M features at the released best epoch (202); and S^{MMEN}, an MLP-based Multi-Modal Embedding Network released as a baseline in [24].

Step 1: row-wise normalization and equal-weight fusion. The three matrices live on different scales (cosine similarities for JPoSE/MMEN, max-margin distances for MI-MM), so a direct sum is biased towards the highest-magnitude model. We min-max normalize each matrix *row-wise* so that, for every video query, scores against the 3,842 candidates are mapped to $[0, 1]$:

$$\widetilde{S}_{ij}^{(m)} = \frac{S_{ij}^{(m)} - \min_k S_{ik}^{(m)}}{\max_k S_{ik}^{(m)} - \min_k S_{ik}^{(m)}}, \quad S^{\mathrm{ens}} = \tfrac{1}{3}\,(\widetilde{S}^{\mathrm{JPoSE}} + \widetilde{S}^{\mathrm{MI\text{-}MM}} + \widetilde{S}^{\mathrm{MMEN}}). \quad (2)$$

Equal weights are the natural default in the absence of a relevance-aligned validation signal.

Step 2: graph-diffusion re-ranking on visual and textual k-NN graphs. Inspired by diffusion-based re-ranking in image retrieval [7] and k-reciprocal encoding [27], we propagate ensemble scores through the local geometry of the retrieval space in both modalities. After L2-normalising each row of S^{ens}, a sparse visual affinity matrix $A_{\mathrm{vv}} \in \mathbb{R}^{N_v \times N_v}$ stores cosine similarity between the score profiles of each pair of segments if either is among the top-k visual neighbors of the other. The same construction on $(S^{\mathrm{ens}})^{\top}$ yields a textual affinity A_{tt}. We then expand the score matrix in both directions and blend with the original ensemble:

$$S^{\mathrm{exp}} = \tfrac{1}{2}\,(A_{\mathrm{vv}}\, S^{\mathrm{ens}} + (A_{\mathrm{tt}}\,(S^{\mathrm{ens}})^{\top})^{\top}), \quad S^{\mathrm{rerank}} = (1-\alpha)\, S^{\mathrm{ens}} + \alpha\, S^{\mathrm{exp}}, \quad (3)$$

with default $k = 20$, $\alpha = 0.3$. Both the ensemble matrix (2) and the re-ranked matrix (3) are submitted to Codabench as separate iterations, isolating the contribution of re-ranking.

4.4 Approach 3: AVION ViT-L fine-tuned with the SMS loss

The third approach replaces inference-time fusion with end-to-end fine-tuning of a strong CLIP-style egocentric backbone, supervised with the soft-label relevance matrix. The visual encoder is the AVION ViT-L/14 of Zhao and Krähenbühl [25], pre-trained on Ego4D via LaViLa [26] and the textual encoder is the CLIP text transformer paired with that AVION checkpoint. Both encoders are fine-tuned jointly with the Symmetric Multi-Similarity loss of Wang, Wang and Chau [19].

Symmetric Multi-Similarity loss. Within a training batch of B video–text pairs, let $\mathbf{x}_i, \mathbf{y}_i \in \mathbb{R}^d$ denote the L2-normalized visual and textual embeddings of pair i, $r_{ij} \in [0, 1]$ the soft relevance between v_i and t_j extracted from R [23], and $s_{ij} = \mathbf{x}_i^{\top} \mathbf{y}_j$. The SMS loss extends the multi-similarity formulation of [20] so that, instead of binary pair weights, the contribution of each pair is modulated by r_{ij}. With margin θ, relevance threshold τ and temperatures β_p, β_n,

$$\mathcal{L}_i^{\mathrm{V}\to\mathrm{T}} = \tfrac{1}{\beta_p} \log\Big(1 + \sum_{j: r_{ij} \geq \tau} r_{ij}\, e^{-\beta_p (s_{ij} - \theta)}\Big) + \tfrac{1}{\beta_n} \log\Big(1 + \sum_{j: r_{ij} < \tau} (1 - r_{ij})\, e^{\beta_n (s_{ij} - \theta)}\Big), \quad (4)$$

and the full loss is the symmetric counterpart $\mathcal{L} = \frac{1}{2}(\mathcal{L}^{\mathrm{V}\to\mathrm{T}} + \mathcal{L}^{\mathrm{T}\to\mathrm{V}})$ averaged over the batch. Compared with InfoNCE-like contrastive losses, SMS pulls together pairs in proportion to their relevance and pushes apart pairs in proportion to $1 - r_{ij}$, providing a supervision signal directly aligned with the nDCG metric.

Fine-tuning configuration. Fine-tuning is launched from the AVION pretrain checkpoint via `ammplus_finetune.py` from a patched fork of the public SMS-Loss codebase [18] on a single A100. Key flags are: `-model CLIP_VITL14`, `-batch-size 48`, `-lr 2e-5`, `-loss-margin 0.6`, `-loss-thres 0.1`, with `-use-fast-conv1`, `-grad-checkpointing` (essential for ViT-L on a single GPU) and `-use-flash-attn` when available. Here, FlashAttention [3] roughly halves attention memory at no accuracy cost. We use AdamW for optimization [11] and one numbered checkpoint is saved per epoch. The relevance threshold $\tau = 0.1$ excludes pairs whose soft relevance is too small to provide a useful gradient and the margin $\theta = 0.6$ controls the separation enforced between positive and negative pairs. Both values are inherited from the public reference configuration of [19]. A trained model can be extended at any point by passing both `-pretrain-model` and `-resume` pointing to the same checkpoint and increasing `-epochs`, we use this hook to extend the base 10-epoch run to 16 epochs.

Divergences from the published recipe. Two configuration choices depart from the SMS-Loss recipe of Wang et al. [19] and shape the upper bound of what our pipeline can achieve. First, our effective batch size of 48 on a single A100 is approximately five times smaller than the 240 samples (60 per GPU $\times$ 4 GPUs) reported for ViT-L by the original authors. Smaller batches reduce the diversity of negatives visible to the SMS denominator in Eq. (4) and weaken the metric-learning signal. Second, our base run is restricted to 10 epochs (extended to 16 in iteration 6) against the 100 epochs of the published recipe. Both deviations are quantified in Section 7.2 and motivate the future-work direction in Section 7.3.

Inference and test-time augmentation. Inference is performed by `test_mir.py`, which encodes all test video segments and text queries and writes a similarity matrix in the same format as Approaches 1 and 2. We compare two settings: a *standard* pass with default 8-frame clips, and a *TTA* pass with `-flip -clip-length 32` that averages the embeddings of the original and horizontally-flipped clips and samples 32 frames per segment. Flip-averaging is the canonical video adaptation of [9] and exploits the near-invariance of kitchen activities to horizontal reflection; the longer clip length is motivated by the long tail of segment durations in Section 3.1.

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5 Results

5.1 Evaluation protocol

All iterations are evaluated under a single protocol on the official EPIC-KITCHENS-100 MIR benchmark on Codabench[1]. Each submission is a Codabench-compatible ZIP containing the 9,668$\times$3,842 similarity matrix serialized as a protocol-2 pickle with a numpy namespace shim. It is scored by the grader against the held-out

[1] https://www.codabench.org/competitions/12008/

Table 1: Official Codabench nDCG AVG (%) across the six iterations of Section 4. Each row attributes a delta to the single design change introduced over the previous iteration. Best per approach in **bold**.

It.	Approach	Single change	nDCG AVG	Δ
1	A1 – JPoSE	Feature-based baseline (no fine-tune)	53.53	–
2	A2 – Ensemble	+ 3-model ensemble (no re-ranking)	55.31	+1.78
3	A2 – Ensemble	+ Graph-diffusion re-ranking (k=20, α=0.3)	**55.82**	+0.51
4	A3 – AVION+SMS	Replace pipeline by AVION ViT-L + SMS, 10 ep	68.80	+12.98
5	A3 – AVION+SMS	+ TTA: `-flip -clip-length 32`	69.48	+0.68
6	A3 – AVION+SMS	+ Six additional fine-tuning epochs (10 → 16)	**69.68**	+0.20
Public leaderboard SOTA, CR-CLIP [6]			82.10	–
Total improvement, It. 1 → It. 6			–	**+16.15**

test relevance matrix [23] and the grader returns the average nDCG (Eq. 1) used to rank submissions. Because the test relevance matrix is not publicly released, we cannot compute mAP or per-direction nDCG offline. Therefore, we report the official Codabench leaderboard nDCG AVG as the single primary metric throughout the paper. As a sanity check on the implementation of Approach 1, the JPoSE codebase exposes an internal train-set validation routine on which we obtained an offline nDCG of 0.690 and mAP of 0.734, matching the figures of Wray et al. [24] before any submission was made. These offline numbers are not directly comparable to the public Codabench score for the same submission (53.53 nDCG) and are therefore not used for cross-iteration comparison.

5.2 Comparative results across iterations

Table 1 reports the official Codabench nDCG AVG for each of the six iterations described in Sections 4.2–4.4, together with the marginal contribution of every design change introduced over the previous iteration. The trajectory closes more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP at 82.10 nDCG [6]) on a single GPU and using only sixteen epochs of fine-tuning.

The breakdown delivers four observations. First, ensemble fusion of three frozen pretrained models contributes +1.78 nDCG, confirming that JPoSE, MI-MM and MMEN are complementary enough for score averaging to recover variance that none of them captures individually, but also that the ceiling of frozen ensembles is modest in this setting. Second, graph-diffusion re-ranking adds +0.51 nDCG on top of the ensemble at no training cost, consistent with the magnitude of the gains historically reported in image retrieval [7]. Third, the single largest jump comes from replacing the inference-time pipeline with a single end-to-end fine-tuned AVION ViT-L supervised by SMS: +12.98 nDCG, dominating every other decision in the iteration plan by an order of magnitude and motivating allocating compute to encoder fine-tuning before exploring fusion strategies. Fourth, post-fine-tuning refinements behave as expected: TTA adds +0.68 nDCG with no training cost and extending fine-tuning from 10 to 16 epochs adds +0.20 nDCG. This suggests the model is approaching the plateau of single-checkpoint training which we revisit in Section 7.3 as motivation for a checkpoint-ensemble strategy.

6 Discussion

The per-iteration breakdown in Table 1 concentrates 80% of the total improvement (12.98 of 16.15 nDCG points) in the single transition from Approach 2 to Approach 3. That is from a frozen-feature ensemble to an end-to-end fine-tuned ViT-L driven by soft-label supervision. The asymmetry along the iteration axis is itself the central observation, and it can be unpacked along three connected lines: the low ceiling of inference-time refinements over frozen backbones, the dominant effect of the encoder-and-loss swap, and the diminishing returns of post-fine-tuning refinements that motivate the future-work direction.

Approach 2 fuses three pre-trained models that were never optimized for the soft-label setting: JPoSE and MMEN are triplet-supervised, MI-MM is binary multi-instance MaxMargin, and none sees the relevance matrix R during training. Equal-weight averaging of their normalized scores recovers +1.78 nDCG on top of JPoSE, in line with the canonical observation that retrieval ensembles reduce variance without correcting bias [22]. Because the three models share the same failure mode (they ignore graded relevance), averaging them produces a more stable estimate of the same flawed objective rather than one that is closer to nDCG. Equal weights were used in the absence of a relevance-aligned validation signal and weighted averaging tuned on a held-out split could plausibly add a fraction of a point but is unlikely to overcome the bias ceiling. Graph-diffusion re-ranking adds another +0.51 nDCG at no training cost, consistent with magnitudes reported for diffusion in image retrieval [7]. A small $k = 20$ keeps the affinity graph sparse and avoids over-smoothing, moderate $\alpha = 0.3$ blends rather than replaces the original signal, and the symmetric construction in Eq. (**??**)—propagating through both visual and textual graphs—prevents either modality from dominating the result.

The single largest jump, +12.98 nDCG between iterations 3 and 4, is driven by three coupled changes that we deliberately bundled into a single iteration boundary by replacing pre-extracted features with an end-to-end ViT-L/14 encoder pre-trained on egocentric video [25,26], replacing binary triplet/MaxMargin objectives by the SMS loss [19] that consumes the soft-label matrix R directly, and jointly fine-tuning both encoders rather than freezing them. Disentangling the relative contribution of each factor would require ablations beyond our compute budget. What is clear is that the gain dominates everything else by an order of magnitude. We attribute most of the effect to the combination of the strong egocentric backbone and the relevance-aware loss. The SMS objective in Eq. (4) weights both positive and negative pairs by their soft relevance, which aligns the gradient signal with the official nDCG metric (Eq. (1)) and is the only mechanism in our pipeline that directly optimises it. The size of the gain is consistent with the observation in [23] that graded-relevance objectives can reshape retrieval orderings well beyond what binary contrastive losses achieve.

Once the encoder has been fine-tuned end-to-end with SMS, however, further gains diminish quickly. TTA (`-flip -clip-length 32`) yields +0.68 nDCG with no training cost. Flip-averaging adapts the canonical recipe of [9] to video and exploits the near-symmetry of kitchen scenes under horizontal reflection, while the longer clip length addresses the long tail of segment durations identified in Section 3.1. Extending fine-tuning from 10 to 16 epochs adds a further +0.20 nDCG, suggesting the model is approaching the plateau of single-checkpoint training. This corresponds precisely to the regime in which model-averaging strategies tend to be most cost-effective and which motivates the future-work direction in Section 7.3. The practical implication of Table 1 is that, under a tight compute budget, encoder fine-tuning with relevance-aware supervision is the highest-yield investment by a wide margin: re-ranking and TTA are essentially free and should be applied by default, ensemble fusion of frozen baselines is also cheap but quickly hits a ceiling, and longer training is subject to rapidly diminishing returns. By contrast, the encoder swap returned +12.98 nDCG for the cost of a single A100 day. This ordering is consistent with what both the SMS-Loss authors [19] and the CR-CLIP authors [6] report on the same benchmark: the largest contribution to their final scores comes from the encoder choice and the relevance-aware loss, with subsequent refinements closing a smaller residual gap.

7 Conclusions, Limitations and Future Work

7.1 Conclusions

We presented a controlled, iterative study of the EPIC-KITCHENS-100 Multi-Instance Retrieval challenge in which three progressively stronger approaches were evaluated under a single Codabench protocol. Starting from a JPoSE feature-based baseline at 53.53 nDCG, a score-level ensemble of three frozen retrieval models combined with graph-diffusion re-ranking lifts performance to 55.82 nDCG, while an AVION ViT-L/14 dual encoder fine-tuned with the SMS loss and complemented by horizontal-flip plus 32-frame test-time augmentation reaches 69.68 nDCG over sixteen epochs of single-GPU training. The final iteration recovers more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP, 82.10 nDCG [6]) using a single A100 instance and the publicly released SMS-Loss training code. The per-iteration breakdown isolates where each gain originates. Encoder capacity coupled with relevance-aware supervision dominates by an order of magnitude, while inference-time strategies contribute small but reliable deltas at zero training cost. Additionally, this provides a transparent compute-allocation guideline for future EK-100 MIR work operating under similar resource constraints.

7.2 Limitations

Two limitations frame our results. First, all Approach 3 experiments were performed on a single NVIDIA A100 (40 GB) with a batch size of 48, which made multi-GPU runs, larger effective batch sizes and multi-seed re-training infeasible within the project timeline. Several knobs that are routine in the EK-100 MIR literature were therefore left unexplored. These include the contribution of larger batches (which directly affects the SMS gradient through the implicit hard-negative mining in Eq. (4)), systematic sweeps over θ, τ, β_p, β_n, longer fine-tuning schedules above 16 epochs, and variance estimation through repeated runs with different random seeds. Second, the EK-100 challenge releases the soft-label relevance matrix only for the training split. As mentioned previously, the test relevance matrix required to compute nDCG and mAP offline is held out by the organizers and is accessible exclusively through the Codabench grader. Model selection during fine-tuning therefore had to rely on training-loss trajectories rather than on a relevance-aligned validation signal, and every reported number in Table 1 required a Codabench submission, which capped the practical iteration budget. As a sanity check on the JPoSE baseline implementation we did obtain offline numbers on the train-set validation routine (0.690 nDCG, 0.734 mAP), but these are not directly comparable to the Codabench score for the same submission (53.53 nDCG) and were not used for cross-iteration comparison.

7.3 Future work

The most promising extension consistent both with the empirical evidence in Table 1 and with the SMS-Loss recipe of Wang et al. [19] is the *score-level ensemble of multiple SMS-supervised checkpoints* described in Section 3.3 of that paper but that we did not have the compute budget to reproduce. Concretely, the published recipe combines six independent models: three Adaptive MI-MM variants with different learning-rate schedules and three SMS variants with different relaxation thresholds ($\tau \in \{0.05, 0.10, 0.12\}$). They are all trained for 100 epochs from the same AVION pretrain [25] and combines them by directly summing their test similarity matrices, achieving an additional +1.1 mAP and +0.6 nDCG over the best single model. Replicating this ensemble on top of our AVION+SMS pipeline is a natural and well-defined extension due to the fact that row-wise normalization and equal-weight averaging used in Approach 2 (Eq. (2)) provide a drop-in replacement for the direct sum, and the graph-diffusion re-ranking of Eq. (3) can be stacked on top at no training cost. The rationale is supported by three observations from our trajectory and the related literature. First, longer training of a single ViT-L instance yields rapidly diminishing returns past 10–16 epochs, so additional compute is better invested in a few short complementary models than in extending a single one indefinitely. Second, model-soup-style strategies—in which several models fine-tuned with different hyperparameters but the same initialization are combined—consistently match or exceed the best individual model in image classification [22] and have already been shown to work in this exact retrieval setting by [19]. Third, and our Approach 2 already validates the inference-time fusion machinery on weak frozen baselines, so deploying it on stronger fine-tuned ViT-L+SMS instances is essentially mechanical. The principal obstacle is computational: training six ViT-L+SMS instances for 100 epochs each requires roughly 40× the GPU-hours of our iteration 6 and is not feasible on a single GPU within a typical project timeline. We therefore frame this direction as the natural continuation of the present work, conditional on multi-GPU infrastructure, in line with the 4×4 RTX 6000 Ada configuration used by Wang et al. for ViT-B and the 48 GB GPUs used for ViT-L. Beyond the ensemble direction, two further extensions follow from the limitations above: integrating the context-refinement mechanism of CR-CLIP [6] on top of our AVION+SMS pipeline (closing a documented portion of the gap to the public leaderboard), and replacing the SMS loss in Eq. (4) with a directly differentiable nDCG surrogate adapted to graded relevance (Smooth-AP [1] or a SoDeep-style ranker [4]).

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